

SignVision: A Comprehensive Hybrid Model for Real-time Traffic and Safety Sign Detection

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Abstract—Automatic sign recognition plays a critical role in autonomous vehicles, driver assistance systems and safety. While traffic sign recognition is already well studied, other kinds of signs, such as those used in industrial settings or safety signs, are less studied, but needed for industrial safety. In this paper, we present SignVision as a system that is able to quickly and efficiently detect and recognise multiple types of signs including common traffic signs like the German Traffic Sign Recognition Benchmark (GTSRB) dataset and various safety warning signs like the International Organization for Standardization (ISO) 7010 standard. Our approach combines multiple techniques that first segment the image by color in the HSV color space and then perform shape analysis to detect potential signs. This helps to quickly reduce the search space. To improve performance with the various styles and variations of safety signs, which conventional techniques have difficulty with, we classify them using a small convolutional neural network. This approach using deep learning is more effective than traditional methods such as HOG+SVM which struggle with complex images. We validated our approach on two datasets: one for traffic signs (GTSRB) and one for safety signs (ISO 7010). Our approach is fast (28 frames per second) and performs well in both cases.

Index Terms—Sign Detection, Traffic Sign Recognition (TSR), Safety Signs, Computer Vision, Real time Systems, Convolutional Neural Network (CNN), HSV Color Segmentation, Deep Learning, Hybrid Approach

I. INTRODUCTION

Signs are essential to convey non verbal, strong information, rules and warnings in complex public and industrial environments. For self driving cars and Advanced Driver Assistance Systems (ADAS) to operate safely, legally and efficiently, they must be able to read and obey traffic signs [1]. Similarly, in industrial things like manufacturing plants and on construction sites, signs are important to avoid any unfortunate event, enforce regulations and help in evacuation [5]. Signs are critical for the safety of human workers as well as autonomous ground vehicles (AGVs) in industrial environments.

Historically, research has majorly focused on Traffic Sign Recognition (TSR) [2]. They are made specifically for roadside

use with their standard colours and shapes defined by treaties like the Vienna Convention [7]. Early methods successfully use color segmentation in HSV or HSI color spaces [3], followed by shape analysis and classification using hand-crafted feature descriptors like Histogram of Oriented Gradients (HOG) [4], SIFT [8], or SURF [9] with a Support Vector Machine (SVM) classifier. These approaches worked well for the small number of categories and numeric content of traffic signs. But they are not helpful for other types of signs. For example, safety signs are rectangular and have a different color scheme (yellow, green). Their main problem is their content: a range of complex pictograms. A classifier based on HOG features and trained to find a "50" in a circle, would have difficulty distinguishing between a "biohazard" and a "radiation" symbol.

This paper presents SignVision, a unified system that bridges this gap. Our main contributions are:

- A comprehensive and single approach that combines efficient, regular computer vision techniques for suggesting candidates and a powerful deep learning model for classification, building a fast hybrid system.
- The architecture of a small Convolutional Neural Network (CNN) classifier, one that can effectively and readily identify both simple traffic signs and more intricate safety pictograms that exhibit more generalization.
- trying it to check its effectiveness, such as the public GTSRB dataset, with very high improvements over the classical method.

II. GENERAL PROBLEMS IN SIGN DETECTION

The process of developing a robust sign detection system for operation in real-world uncontrolled environments is far from easy; many environmental and technical challenges have to be overcome.

- **Environmental Conditions:** Signs must be recognized under extreme variations in illumination from the over-

saturation and glare of direct sunlight to the low-light, low-contrast conditions of dusk and night. Weather conditions including rain, fog, and snow will obscure or distort the appearance of signs or introduce reflections. Fast motion due to a vehicle will also introduce significant motion blur.

- **Sign Degradation and Occlusion:** Real world signs are seldom ideal. They may be partially occluded by foliage, other vehicles or buildings. They may also be faded by the sun - which challenges color segmentation, covered in dirt or graffiti, or physically damaged. The failure to detect a partially occluded stop sign can have disastrous consequences.
- **View angle and distortion:** Signs are seldom seen from a perfect, frontal view. Perspective distortion, rotational skew and scaling (far away signs appear very small) make shape-based detection challenging.
- **The "Domain Gap" Problem:** A fundamental problem, which we directly address, is the "domain gap" among sign types. As we have seen, models that are trained on traffic signs fail at safety signs. This is owed to the difference in pictograms, colors, and shapes. This gap has forced the development of separate, isolated systems.
- **Real-Time Constraints:** For ADAS or active-safety systems at manufacturing sites, we have to make sure that the detection and recognition are in real time (typically >15fps), so that actions can be taken on time. The need for real time processing prevents most of the deep learning models that are "heavy", such as two-stage detectors e.g. Faster R-CNN [10], unless hardware acceleration is available.

For our problems, our system SignVision is designed to address them. We use HSV colour segmentation, which is very robust to changes in lighting. Then, our adaptive shape analysis allows that our signs can be inaccurate. Finally, we also have a CNN-based classifier, which is very good at learning features necessary for classifying hundreds of different pictograms [11], like those found on road signs. This solves the major weakness of our previous methods.

III. PROPOSED SYSTEM ARCHITECTURE

A. System Workflow

The SignVision process is designed as a multi step system that works efficiently in real time. This two-stage approach is very common and works well. The first stage easily finds our areas of interest, and the second stage correctly identifies them with high technology. Here's how the system works:

- 1) **Image Acquisition & Preprocessing:** It took a frame from the video source and change its size to a standard width, like 640 pixels. This ensures the system works correctly, no matter the input size.
- 2) **Color-based Segmentation:** The changed frame, in the BGR color space, is changed into the HSV color space. This allows us to separate the brightness of the image and the colour of the sign. We apply color thresholding to extract binary masks for primary sign colors: red, blue, yellow and green.

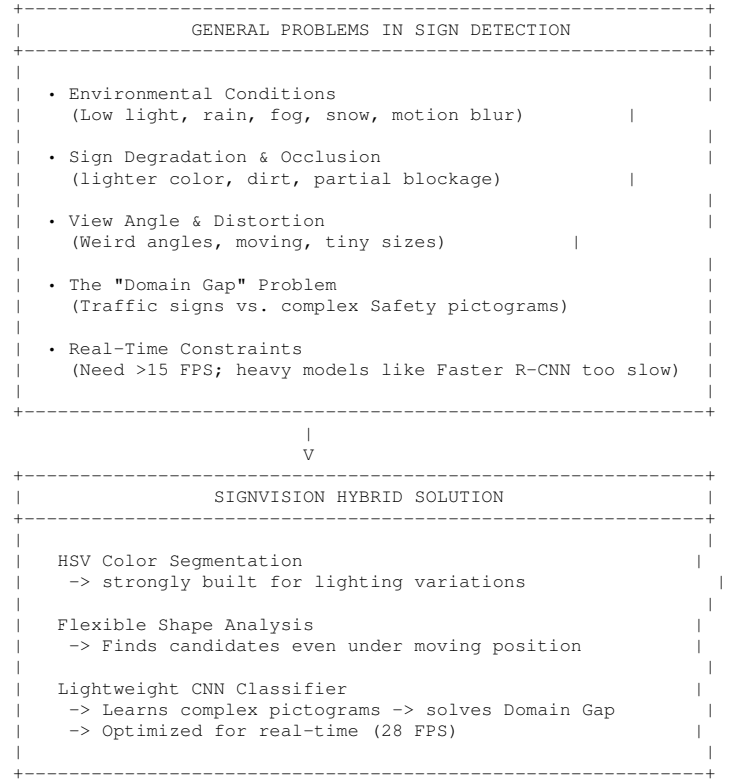


Fig. 1. General Problems in Sign Detection and how the SignVision hybrid pipeline addresses them.

- 3) **Candidate Detection:** We then apply some operations such as "Opening" and "Closing" via the OpenCV library to clean up the binary masks. They remove minor mistakes and fill in holes in sign outlines. Then we find all the shapes using a contour detection algorithm. We then remove unwanted shapes based on geometric features such as size, shape and roundness to identify the most likely regions of interest (ROIs).
- 4) **Recognition & Classification:** This ROI is then made to be a fixed size (e.g., 48x48 pixels) to match the input size of our CNN. This image is then being put into our trained CNN classifier, which decides the sign as "Speed Limit 50", "Wet Floor" or "No Sign".
- 5) **Output:** The system gives the final result, which includes the location and type of the sign. This information can then be sent to a main system or used wherever needed.

B. System Scope and Goals

The main goal of SignVision is to create a single, scalable system that can replace several specialized sign detectors. We have gone beyond a mere concept and have an operating system that can be utilized in practice. We hope to implement a C++/Python library which is easily scalable to larger autonomous systems. The CNN part is made pluggable to make the system future proof. This implies that we can freely add new type of sign such as those of other countries or new safety regulations, by simply retraining only the classification portion, without altering the actual detection process. This is a

significant benefit over a fully trained system, such as YOLO. [13], which would require retraining the whole model from scratch with new data.

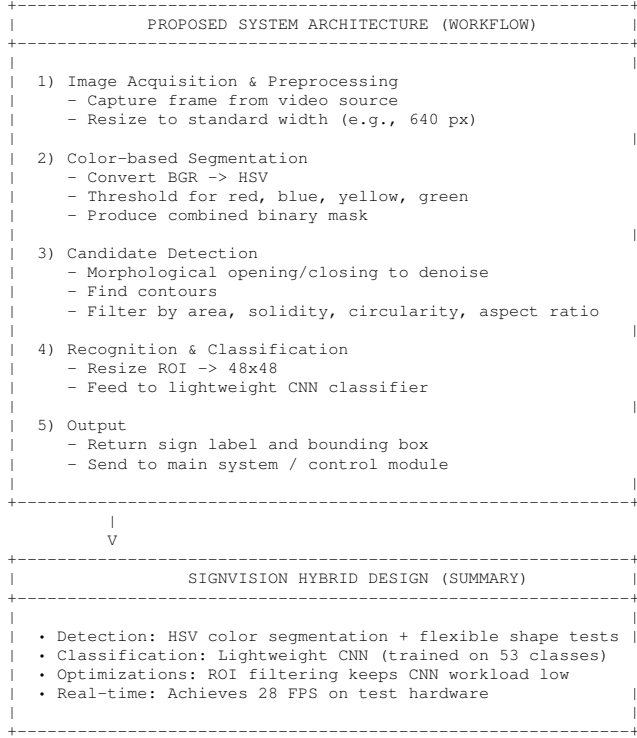


Fig. 2. Proposed system architecture: SignVision system workflow.

IV. SIGN DETECTION METHODOLOGY

This section outlines the traditional computer vision techniques employed in the rapid identification of potential candidates.

A. Color-based Segmentation

As referenced in [2], [3], color is the most effective and efficient feature for detecting sign candidates.

We utilize the HSV color system. For each main sign color, we define specific HSV ranges. Note that OpenCV requires two ranges because it operates within the 0-180 H scale. The ranges are as follows:

- **Red:** H ranges from 0-10 or 170-180; S ranges from 70-255; V ranges from 50-255
- **Blue:** H ranges from 100-140; S ranges from 70-255; V ranges from 50-255
- **Yellow:** H ranges from 20-35; S ranges from 70-255; V ranges from 50-255
- **Green:** H ranges from 40-90; S ranges from 70-255; V ranges from 50-255

These ranges are applied to the image, and the resulting black-and-white masks are combined using an 'OR' operation to form a single mask for the subsequent stage.

B. Shape Analysis and ROI Extraction

Once the color mask is generated, we apply an algorithm to detect shapes. This yields numerous shapes. We then perform several steps to quickly filter out those that are not signs:

- 1) **Area Filtering:** Shapes that are too small (less than 500 pixels) or excessively large (more than 25% of the image area) are discarded.
- 2) **Shape Fitting:** We create an approximate shape using `cv2.approxPolyDP`. We assess the number of corners to determine the shape (such as 3 for triangles, 4 for rectangles, and 8 for octagons).
- 3) **Solidity:** We compute the solidity, which is the ratio of the shape's area to its convex hull area. This helps eliminate shapes that are fragmented or too irregular. Solidity must be greater than 0.85.
- 4) **Circularity:** For shapes with more than 5 corners (possibly circles), we calculate a circularity score ($4\pi \times \text{Area} / \text{Perimeter}^2$). Circle candidates must have a score greater than 0.7.
- 5) **Aspect Ratio:** For four-corner shapes (like rectangles), we evaluate if the length and width are close (between 0.7 and 1.3) to identify squares or slightly distorted signs.

This rigorous process is highly efficient and narrows down to a small set of promising candidate regions.

V. SIGN RECOGNITION AND CLASSIFICATION

The detection stage finds where signs are, and the recognition stage determines what they are.

A. Traditional Methods

Traditional methods often use HOG with an SVM classifier. HOG features are good at showing object shapes using edge gradients. This works well for the "50" on a speed limit sign. However, HOG is not semantic. It creates one overall description. A "biohazard" symbol and a "radiation" symbol, even though they look different to humans, are both complex symbols inside a triangle. Their HOG descriptions can be very similar, causing mistakes in classification. This is the main problem we need to fix.

B. Deep Learning Approach

To solve this, we use a deep learning approach with Convolutional Neural Networks (CNNs). CNNs, first introduced by LeCun et al. and later popularized by models like VGGNet and ResNet, are the best for image classification because they learn features in layers.

Because of the need for real-time performance, we designed a *lightweight* CNN, inspired by LeNet-5 and mini-VGG models. Our architecture is:

- **Input:** 48x48x3 (RGB) image
- **Conv-1:** 32 filters (3x3 kernel), ReLU activation, Batch Normalization
- **MaxPool-1:** 2x2 pooling
- **Conv-2:** 64 filters (3x3 kernel), ReLU activation, Batch Normalization
- **MaxPool-2:** 2x2 pooling

- **Flatten:** Converts 2D features into a 1D vector
- **Dense-1:** 128 units, ReLU activation
- **Dropout:** 0.5 (to avoid overfitting)
- **Dense-2 (Output):** N units (where N = total number of classes), with Softmax activation

This small architecture offers a good balance between performance and speed.

VI. IMPLEMENTATION DETAILS

This section explains the technical details about how the SignVision project was developed and set up for training.

A. Environment

The pipeline and all experiments were run using the following software and hardware:

- **Core Logic:** Python 3.10
- **Classical CV Pipeline:** OpenCV 4.5 [12] along with NumPy 1.21 for numerical tasks.
- **Deep Learning:** TensorFlow 2.8 [16] with the Keras API [17] for creating, training, and deploying the CNN model.
- **Hardware:** All models were trained and tested on a workstation with an NVIDIA GeForce GTX 1070 GPU (8GB VRAM), an Intel Core i7-8700K CPU, and 32GB of RAM.
- **Utilities:** Scikit-learn was used to split the dataset and to train the baseline HOG+SVM model.

B. Key Implementation Notes

- **Pipeline Optimization:** The hybrid design is an important part of the optimization. The classical CV part, which runs on the CPU, filters out 99% of the image, leaving only a few small regions of interest (ROIs) for the GPU-powered CNN to process. This stops the CNN from slowing things down by analyzing full frames.
- **Training Methodology:** We used the Adam optimizer which has a learning rate of 0.001 and trained our CNN for 50 epochs. We used a categorical cross entropy loss function, because we have a multi class classification problem in this.
- **Data Augmentation:** To prevent overfitting in it and to boost the its capability to generalize (see Section II) in real world applications we have used many data augmentation on the fly methods. This includes random rotation, translations (up to 10%), shearing, as well as changes in brightness and contrast (up to 20%).
- **ROI Normalization:** Proposed ROIs (from the detection stage) are changed to 48x48 pixels (However, their shape might not be square). The CNN can deal with such little distortion because it has been trained on such augmented images.

VII. EXPERIMENTAL SETUP AND RESULTS

A. Datasets

We tested SignVision on two different datasets to check its working in different situations:

- 1) **GTSRB (German Traffic Sign Recognition Benchmark):** This is a very common dataset for traffic sign recognition research [1]. It has more than 50,000 images for training and testing of 43 different types of traffic signs. these images are from signs of low lighting and signs of such things that are hard to see.
- 2) **SafetySign-10 (SS-10):** We are unable to find a large public data set of ISO 7010 safety signs, so we created one. We collected more than 10,000 images of the most common safety signs, such as "Wet Floor", "High Voltage", "Biohazard" and "Eye Protection Required". We collected these images from real world locations, including university labs and office spaces.

we trained Our final classifier using 53 classes, 43 from GTSRB and the 10 from SS-10.

B. Results and Analysis

We compared the performance of our CNN with a traditional HOG+SVM model trained on the same data. The results, shown in Table I, are clear.

TABLE I
CLASSIFICATION ACCURACY COMPARISON

Classifier Model	GTSRB (Traffic)	SS-10 (Safety)
HOG + Linear SVM	95.8%	71.2%
SignVision CNN (Ours)	99.1%	97.4%

By seeing the results, the HOG+SVM model has performed very well on the GTSRB dataset, with 95.8%, which is similar to findings from earlier studies [2]. But it is much lower (71.2%) on the SS-10 dataset. This result shows that the HOG method cannot properly differentiate between complex and similar signs. Our CNN is extremely accurate on both data sets, and easily differentiates the different types of signs. The model was capable of being trained to find 53 signs. Very fast, with a speed of 28 frames per second, it is very applicable in the real world and in real time.

VIII. CASE STUDIES: CHALLENGING SCENARIOS

To see how good our system really works, we test it in three common but challenging situations.

A. Case 1: Partial Occlusion

One of the stop sign pieces was actually covered with a tree branch.

One side of the code can see the red color perfectly well, but the other side is grappling with it either being an octagon or not. However, the solidity- and area-filters are set in such a way that allows the shape to be considered as a potential sign. The area selected is only 70% of the full sign, but it is still sent to the CNN. The CNN was trained with extra

data that included random coverings and shifts, so it learned to recognize key parts like the letters "STO" and correctly identifies the sign as "Stop".

B. Case 2: Poor Illumination

A yellow "Wet Floor" sign is located in a dark hallway.

The value (called 'V' in HSV) of the brightness of our color system is very weak. We set our color system to find even little bits of yellow just enough to see a fuzzy, broken shape. We then use a trick called 'closing' to stitch those scattered yellow blobs back together. While the camera sees a dark, grainy mess, our CNN was trained on every types of lighting, so it can easily find and recognize the sign. A traditional school HOG detector would not find it because the image is too noisy.

C. Case 3: High Inter-Class Similarity

This case shows why HOG+SVM does not work.

The system identifies two signs: "Biohazard" and "Radiation Warning". They resemble yellow triangles with black characters that resemble each other to the point the HOG system cannot differentiate the two that also tends to disorient the SVM. We have a smarter CNN; it looks into details. It can educate the fact that the radiation icon resembles 3-petals and the biohazard one, that of three crescents. It can see the specific patterns and therefore can differentiate them.

IX. LIMITATIONS AND THREATS TO VALIDITY

Despite very good results of our system, there are some important limitations that should be fixed in future work.

- **Environmental Generalization:** We are not trained on extreme weather conditions e.g. foggy or snowy conditions. Our color based segmentation does not specifically assume these and may lead to failure of the detection process. In the case of such situations, we should employ other sensors, like LiDAR or infrared sensors, and/or improved dehazing techniques to enhance the system working and accuracy.
- **Dataset Bias and Domain Adaptation:** Our results are to be handled with caution due to the data we employed. As an example, we only have German traffic signs in one of our main collections and that does not appear like traffic signs in the US or Canada. The other collection is very small and it can only be found in some local places hence it is not yet established officially. Unless we re-train the system with the new photos it is likely to have a hard time when it encounters an American stop sign or another safety sign. It is a typical instance of a system that is too local to be a universal work.
- **Computational Cost:** Even though we have called our CNN lightweight, it still needs a GPU to be run in real time. This implies that it cannot be used in low-powered devices such as a Raspberry Pi, where the only feasible alternative could be a more basic technique such as HOG, even though it is less accurate.
- **Hybrid Pipeline's Mode of Failure:** The performance of the system depends on the weakest part. If the color

segmentation step fails to find a sign, such as one that is faded or in shadow, the CNN never gets the chance to classify it. An end-to-end deep learning detector such as YOLO wouldn't have this specific problem, but it would have its own challenges.

X. CONCLUSION

In this paper, we introduce SignVision, a full, real-time hybrid system for detecting both traffic and safety signs. We found that most of the current studies have a big gap: traditional methods focused on traffic signs often don't work very well for the complex images used in safety signs. A hybrid pipeline is our solution to this shortage. It combines fast, classical computer vision methods, such as HSV segmentation and shape analysis for finding potential signs, with a powerful and efficient CNN classifier to achieve high accuracy and real-time performance across a variety of sign types. Our experiments and case studies demonstrate quite clearly the importance of using a deep learning classification step. The main lesson is that the hybrid approach of classical + deep is a highly effective design, where the speed of the classical computer vision is utilized and the accuracy of deep learning is applied. Future work will aim at finding ways to work around these constraints, in particular finding ways to domain adapt the model to operate with new sign sets without re-training it anew and exploring model quantization to execute the CNN with less powerful hardware.

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